

sizing instances and using reserved instances can significantly reduce costs.

There are quite a few benefits of operational efficiency in cloud DevOps.

Faster time-to-market: By automating and streamlining processes, organisations can release new features and updates more quickly, gaining a competitive edge.

Improved quality and reliability: Continuous testing and monitoring ensure that issues are detected and resolved early, leading to more stable and reliable applications.

Cost savings: Efficient use of resources and effective cost management practices reduce unnecessary expenditure, optimising the overall IT budget.

Enhanced collaboration: DevOps fosters a culture of collaboration between development and operations teams, breaking down silos and improving communication and productivity.

Scalability and flexibility: Cloud-based DevOps solutions provide the ability to scale resources dynamically, ensuring that the infrastructure can handle varying workloads efficiently.

Popular third party DevOps solutions

Some of the most popular third party DevOps tools today are briefly outlined here.

→ Jenkins

Pros

- Highly extensible with a vast library of plugins.
- Strong community support.
- Free and open source.

Cons

- Complex setup and maintenance
- Can become resource-intensive

Technical features

- **CI/CD automation:** Automates the process of integrating code changes, testing, and deploying applications.

- **Pipeline as code:** Jenkins file allows defining build, test, and deployment stages in code.
- **Distributed builds:** Supports running builds on multiple nodes, distributing the workload to optimise build times.
- **Extensible plugin architecture:** Thousands of plugins available to integrate with various tools and platforms (e.g., Slack, Docker, AWS).
- **Declarative and scripted pipelines:** Provides both declarative (simpler, structured) and scripted (flexible, powerful) pipeline syntaxes.
- **Integration with SCM:** Supports various source control management (SCM) systems like Git, SVN, Mercurial, and more.
- **Built-in security:** Role-based access control, matrix-based security, and integration with LDAP or Active Directory.

→ Gitlab CI/CD

Pros

- Integrated with GitLab repository management.
- Comes with built-in CI/CD pipelines.
- Has robust security features.

Cons

- Can be overwhelming for beginners.
- Limited to GitLab repositories unless using external SCM integration.

Technical features

- **Auto DevOps:** Automatically detects, builds, tests, and deploys applications based on project structure.
- **CI/CD pipelines:** Integrated with GitLab repositories, offering seamless automation from code commit to deployment.
- **Kubernetes integration:** Native support for deploying applications to Kubernetes clusters.
- **Container registry:** Built-in Docker container registry to store and

manage container images.

- **Security scanning:** Includes SAST (static application security testing), DAST (dynamic application security testing), and dependency scanning.
- **Review apps:** Automatically generates temporary environments for review purposes for each merge request.
- **Pipeline visualisation:** Graphical representation of pipeline stages and jobs, providing easy-to-understand insights into the pipeline flow.

→ CircleCI

Pros

- Easy to set up and use.
- Highly scalable.
- Extensive integration options.

Cons

- Limited free plan.
- Some advanced features require a paid plan.

Technical features

- **CI/CD pipeline automation:** Automates the build, test, and deployment processes with customisable workflows.
- **Docker support:** Native support for building and deploying Docker containers, with pre-built images available.
- **Orbs:** Reusable packages of CircleCI configuration for commonly used tasks, simplifying pipeline configuration.
- **Parallelism:** Splits tests across multiple containers to speed up test execution times.
- **Matrix jobs:** Runs multiple job configurations in parallel, useful for testing against different environments.
- **Caching:** Caches dependencies and other build artifacts to speed up subsequent builds.
- **Integration with VCS:** Supports GitHub, Bitbucket, and GitLab repositories for seamless code integration.